Exploring Gene Expression in Breast Cancer Samples from the TCGA-BRCA Cohort

Jewel Irvin

October 19, 2026

1 Introduction

Breast cancer is a heterogeneous disease characterized by biological differences among tumors. Gene expression analysis can help describe these differences and explore patterns associated with molecular subtypes. This project examines gene expression in the TCGA-BRCAcohort, focusing on ESR1 as the primary gene and ERBB2 as a secondary gene. PAM50 molecular subtype information is used to examine how ESR1 expression varies across breast cancer subtypes.

1.1 Research Question

How does ESR1 gene expression vary across breast cancer samples and PAM50 molecular subtypes?

2 Methods

2.1 Data Preparation

The analysis uses a gene-expression count matrix and a metadata table. The count matrix contains gene identifiers and expression counts across samples and the metadata table contains PAM50 molecular subtype annotations.

ESR1 and ERBB2 were selected for analysis using Ensembl ID.

2.2 Summary Statistics

For ESR1 and ERBB2, the mean, median, standard deviation, minimum, and maximum expression counts were calculated. These statistics summarize the distribution of observed counts for each gene.

Gene	Mean	Median	SD	Minimum	Maximum
ESR1	31442.22	17561	38261.25	14	270977
ERBB2	46982.49	19125	105287.55	205	995067

3 Results

3.1 ESR1 Expression Distribution

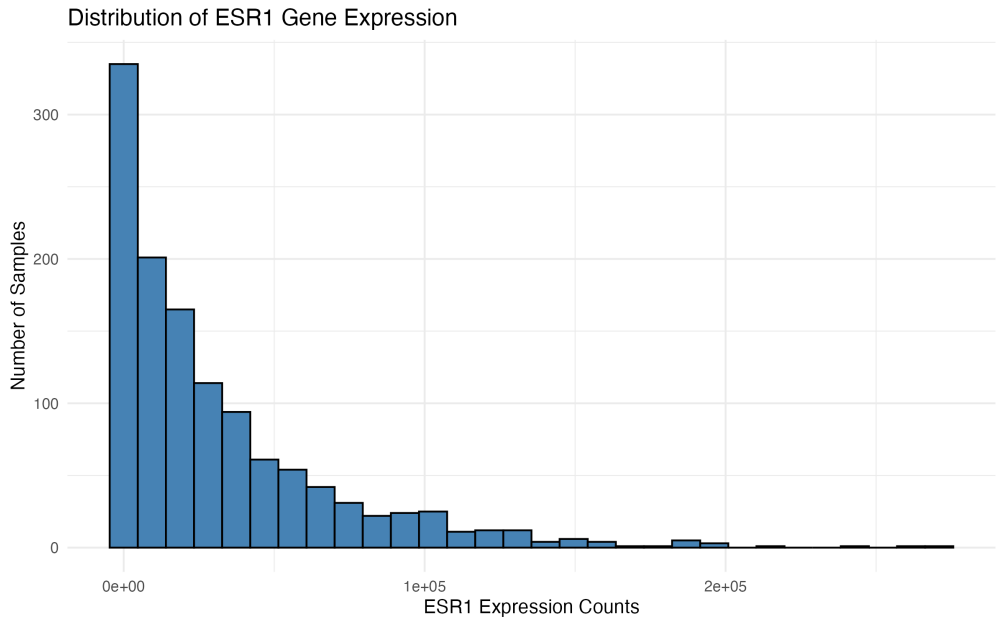


Figure 1: Distribution of ESR1 gene expression counts across TCGA-BRCA samples.

The histogram illustrates the distribution of ESR1 gene expression counts across breast cancer samples. The distribution is right-skewed, with most samples exhibiting lower expression counts and a smaller number showing substantially higher counts.

3.2 ESR1 and ERBB2 Expression

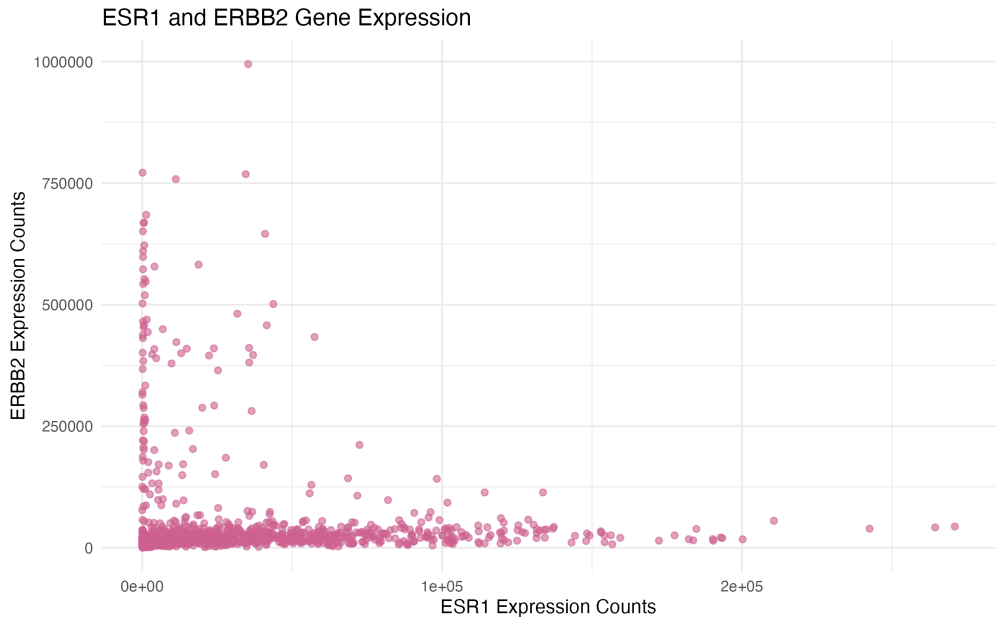


Figure 2: ESR1 and ERBB2 gene expression across breast cancer samples.

The scatter plot examines the relationship between ESR1 and ERBB2 gene expression across breast cancer samples. Each point represents a sample, with its position indicating the expression counts of the two genes. The plot shows variation in expression levels across samples, including some with relatively high ERBB2 expression.

3.3 ESR1 Expression by PAM50 Subtype

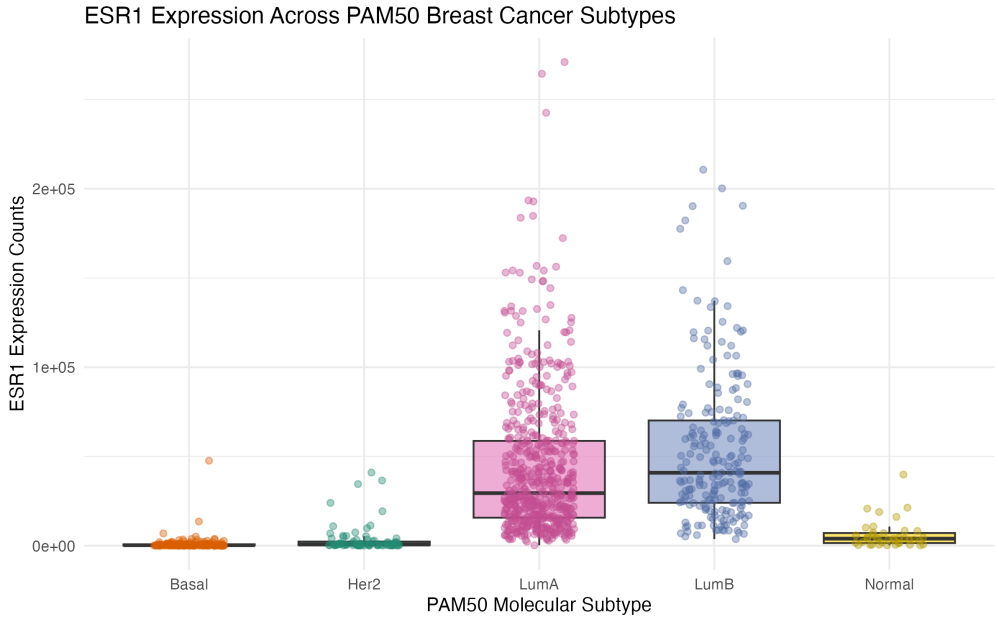


Figure 3: ESR1 expression across PAM50 molecular subtypes.

The box plot compared ESR1 expression across the five PAM50 molecular subtypes: Basal, HER2-enriched, Luminal A, Luminal B, and Normal-like. The medians, interquartile ranges, whiskers, and potential outliers provide a visual comparison of expression levels and variability across the groups.

3.4 Expression Patterns Across Ten Genes

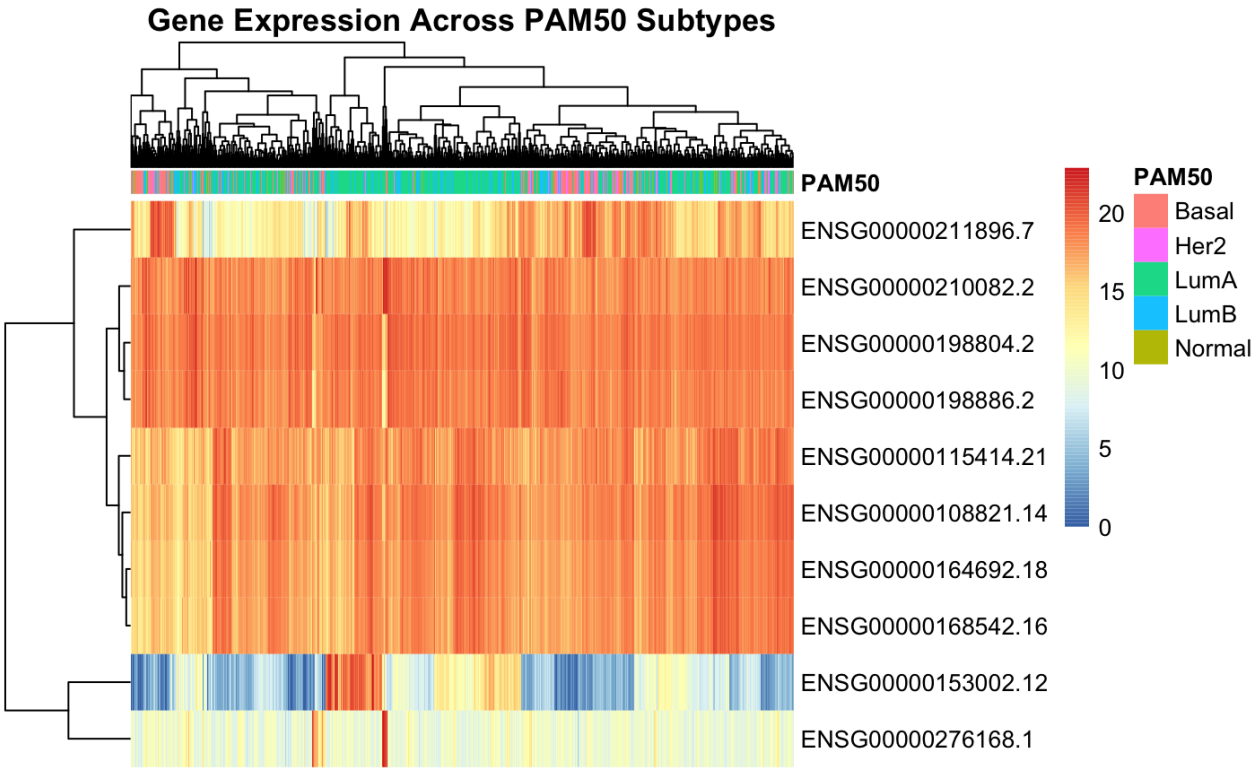


Figure 4: Expression patterns across ten selected genes in the TCGA-BRCA cohort.

The heatmap visualizes gene expression patterns for ten selected genes across breast cancer samples, with PAM50 molecular subtype annotations. Differences in color represent differences in expression counts, allowing patterns across genes and samples to be compared.

3.5 Distribution of PAM50 Subtypes

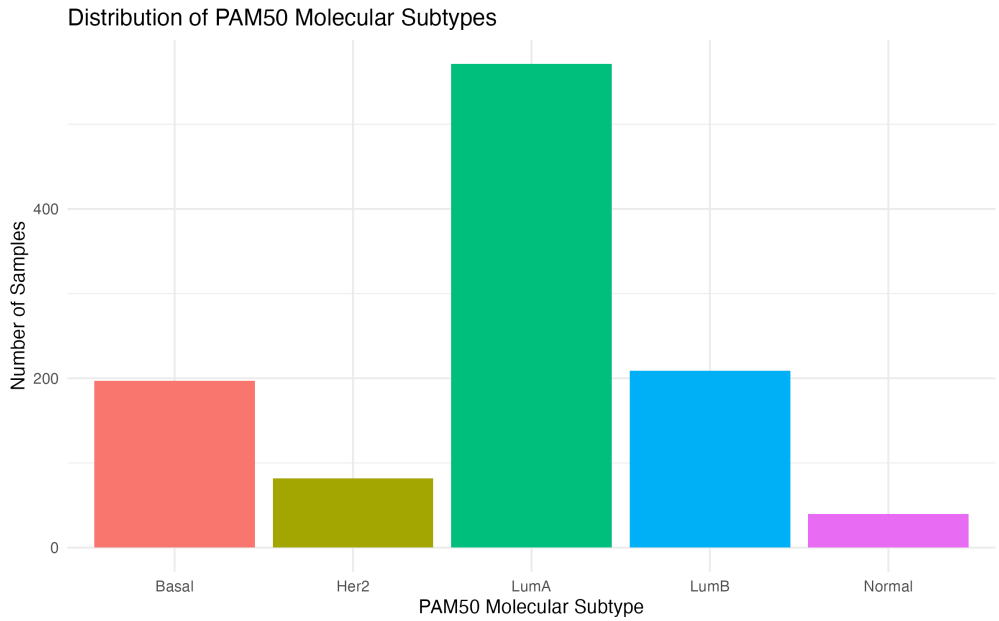


Figure 5: Number of samples in each PAM50 molecular subtype.

The bar chart displays the number of breast cancer samples belonging to each PAM50 molecular subtype. The distribution shows that the subtype groups are unevenly represented in the dataset. This imbalance is important to consider when comparing gene expression across subtypes because estimates for groups with fewer samples may be less stable.

4 Conclusion

Ultimately, this analysis explored how ESR1 gene expression varies across breast cancer samples and PAM50 molecular subtypes in the TCGA-BRCA cohort using five visualizations. The histogram revealed a right-skewed distribution of ESR1 expression, highlighting the importance of understanding the distribution and variability of ESR1 expression across samples. The scatter plot compared ESR1 and ERBB2 expression, revealing wide variation in the expression levels of both genes across samples and allowing for the exploration of a potential relationship between them, although further statistical analysis would be needed to determine the strength and direction of any association. The box plot compared ESR1 expression across the five PAM50 molecular subtypes, providing a visual method for examining differences in median expression, spread, and potential outliers between groups. The heatmap displayed expression patterns across ten selected genes and samples, helping identify similarities and differences in gene-expression profiles and providing a broader view beyond ESR1 alone. Finally, the bar chart illustrated the distribution of molecular subtypes in the cohort. Luminal A was the most represented subtype, with 571 samples, while Normal-like was the least represented, with 40 samples. Together, these visualizations provided complementary perspectives on gene-expression distributions, relationships between selected genes, differences across molecular subtypes, and cohort composition.

5 References

1. OpenAI. (2026). *ChatGPT*. Used for R coding assistance and troubleshooting. https://chatgpt.com/

2. Course instructor. (2026). *counts.csv*. Gene-expression count matrix provided for the course project.

3. Course instructor. (2026). *meta_data.csv*. Sample metadata file provided for the course project.